

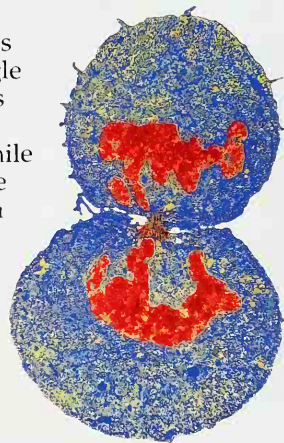


DECIPHERING THE GENETIC CODE. Genes are made of a chemical named deoxyribonucleic acid, or DNA. This has a corkscrew-shaped (helical) double molecule. Each helix carries rows of chemicals known as bases, and their varying order determines the genetic instructions. The structure was discovered by American biologist James Watson (1928-) and English biochemist Francis Crick (1916-). With Maurice Wilkins, they received a Nobel Prize in 1962.

Growth and development

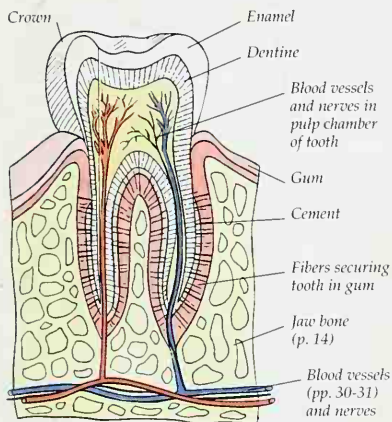
NEVER AGAIN DOES THE HUMAN BODY grow as fast as it did in the womb (pp. 44-45). The single cell of the fertilized egg has multiplied billions of times. If growth continued at the same rate after birth as before, a baby would be over 1 mile (2 km) tall by its first birthday. Throughout the first 20 years of life, the body's general pattern of growth is headfirst. The brain and head lead the way, followed by the chest and abdomen, then arms and legs. A new baby has a relatively large head and short limbs. The torso catches up during childhood, and the arms and legs are the last to lengthen during the teenage years. This pattern of growth, like all of the body's physiological

events, is controlled by genes. These are instructions, coded in a complex chemical form, for building and maintaining the structure and chemistry of a human being. There are 50,000-100,000 genes for a human body. Every tissue cell, such as bone, fat, or muscle, has two copies of all of them, held in the nucleus (pp. 12-13). The cells involved in reproduction – the gametes, or egg and sperm cells – have only one copy so that when they unite at fertilization, the pair is made. Red blood cells have no nucleus, and therefore do not carry genes.



CELL DIVISION

Growth is mainly about making more cells, not bigger ones. Cells not directly involved in producing gametes for reproduction (pp. 42-43) multiply by dividing into replicas of themselves. This is known as mitosis. After a period of cell growth the genetic material, in red in this colored electron micrograph, duplicates. The two sets of genes move apart. The outer membrane constricts, and the cell pinches itself into two separate cells.



INSIDE A TOOTH

The crown of the tooth, exposed above the gum, is anchored by long roots, glued by living "cement" into the jaw bone. Under the incredibly hard enamel is slightly softer, shock-absorbing dentine, to make chewing more comfortable. Within this is the pulp chamber, containing blood vessels that nourish the living tooth, and nerves (pp. 58-59) that warn of injury, disease, or decay.

Teeth

Teeth are specialized structures originating from bone, which undergo growth and replacement during early life. They are covered with the hardest substance in the body, enamel, which has to withstand almost a lifetime of cutting and grinding. The four different types of teeth have different roles, though these are not as marked as in mammals such as dogs and horses. Chisel-shaped incisors cut and slice; pointed canines grip and tear; and broad, flat premolars and molars crush and chew.



BIRTH DAY TEETH

A newborn baby's teeth are usually hidden below the gums. They become visible, or erupt, from the age of about six months.

FIVE-YEAR TEETH

The first 20 teeth to grow are called the primary, deciduous, or milk teeth. From about six years these begin to fall out, and the adult set grows in their place.

FULL SET OF ADULT TEETH

By 21 years of age, all 32 teeth are usually through. In each side of each jaw are two incisors, one canine, two premolars, and three molars. In some people the third molars never erupt.